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United States Patent	12400225
Kind Code	B2
Date of Patent	August 26, 2025
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Systems and methods for beacon and cloud payments

Abstract

Disclosed herein are systems and method comprising: transmitting, by a point-of-sale system, a beacon identifier receivable by a mobile device; and receiving, by the point-of-sale system and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for the product. Further disclosed are systems and methods comprising: receiving, at a mobile device, a beacon identifier from a point-of-sale system, the beacon identifier associated with a product; and transmitting, by the mobile device and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for the product.

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Appl. No.: 15/445284

Filed: February 28, 2017

Prior Publication Data

Document Identifier	Publication Date
US 20180247307 A1	Aug. 30, 2018

Publication Classification

Int. Cl.: G06Q20/40 (20120101); G06Q20/20 (20120101); G06Q20/32 (20120101); G06Q50/06 (20240101)

U.S. Cl.:

CPC **G06Q20/40** (20130101); **G06Q20/202** (20130101); **G06Q20/3223** (20130101);
G06Q20/3224 (20130101); **G06Q20/327** (20130101); G06Q50/06 (20130101)

Field of Classification Search

CPC: G06Q (20/322); G06Q (20/327); G06Q (20/3278); G06Q (20/401); G06Q (20/405);
G06Q (20/10); G06Q (20/20); G06Q (20/3224)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0110296	12/2012	Khoo	700/286	B60L 53/31
2014/0006188	12/2013	Grigg	705/17	G06Q 20/3276
2014/0335897	12/2013	Clem	455/456.3	G08G 1/147
2015/0170135	12/2014	Fourez	705/21	G06Q 20/40
2015/0227928	12/2014	Patel	705/71	G06Q 20/32
2015/0228017	12/2014	Momin	705/38	G06Q 40/03
2016/0182170	12/2015	Daoura	455/3.01	H04W 12/033
2017/0061511	12/2016	Korra	N/A	H04W 4/029
2017/0076265	12/2016	Royyuru	N/A	G06Q 20/18
2017/0114958	12/2016	Otaki	N/A	F17C 13/026
2017/0127230	12/2016	Enriquez	N/A	H04L 67/18
2018/0114217	12/2017	Kohli	N/A	G06Q 10/109

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013222118	12/2013	AU	G06Q 20/204
WO-2016082687	12/2015	WO	G07B 15/06

OTHER PUBLICATIONS

<https://www.paymet.com/gas-station-payments.html> (Year: 2016). cited by examiner

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Background/Summary

BACKGROUND

(1) Currently, in order to pay for something a consumer must retrieve a form of payment and interact with either a cashier or other self-service terminal. For example, when a consumer wants to purchase fuel for his or her automobile, he or she must interact with the pump or go into the store to first pay for the fuel. In a retail environment the consumer or a cashier is required to scan or otherwise enter product information into a point-of sale (POS) system and the consumer must produce a form of payment to be entered into the POS system as well.

SUMMARY

(2) Disclosed herein are systems and method comprising: transmitting, by a point-of-sale system, a beacon identifier receivable by a mobile device; and receiving, by the point-of-sale system and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for the product. Further disclosed are systems and methods comprising: receiving, at a mobile device, a beacon identifier from a point-of-sale system, the beacon identifier associated with a product; and transmitting, by the mobile device and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for the product.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 shows an operating environment for consistent example embodiments disclosed herein;
- (3) FIG. 2 shows an example schematic of a computing device consistent with example embodiments disclosed herein; and
- (4) FIG. 3 shows an example flowchart for a method consistent with example embodiments disclosed herein.
- (5) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary embodiments of the invention, and such exemplifications are not to be construed as limiting the scope of the invention any manner.

DETAILED DESCRIPTION

- (6) The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While embodiments and examples are described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements and stages illustrated in the drawings, and the systems and methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods or elements to the disclosed systems. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of any invention disclosed herein is defined by the appended claims.
- (7) Traditional point-of-sale (POS) systems require that a customer produce and physically interact with a tangible form of payment (e.g., cash, credit card, check, smart phone for Apple Pay®, etc.) in order to purchase goods or services. This is inefficient. For example, a traditional fuel pump requires that a user produce a credit card or go inside and pay with a tangible form of payment before the driver can dispense fuel. During busy times (e.g., rush hour), long lines can occur at the pumps. Lines can get even longer when customers who are filling up go inside the store and overestimate how long it will take to fill their gas tanks because they do not realize the tanks are full and move their cars so others can use the pumps. This wastes customers' time while simultaneously hurting the business because idle pumps represent lost revenue.
- (8) As disclosed herein, POS systems may be outfitted with a beacon. The beacon can interact with mobile devices such as, but not limited to, smart phones, smart watches, smart cars, etc. The POS systems also may be connected to a network, or cloud system. The beacon and network or cloud system may allow the POS system to communicate with the mobile device. The communications between the POS system and the mobile device may greatly increase efficiency by allowing mobile device and POS system to interact with one another without any direct input from the user. Stated

another way, the transaction can occur in the background without the consumer having to interact with the POS system, cashier, or otherwise produce a tangible form of payment.

(9) Using a fuel pump as an example, a user may approach the pump and a beacon connected to the pump may alter the user's mobile device that the user is at a gas station. The user's mobile device can then transmit, via the cloud, payment information to the pump without the consumer having to produce a credit card, press a button on his or her mobile device, or otherwise interact with the pump other than placing the filling nozzle in the automobile.

(10) As disclosed herein, the POS system and the mobile device can exchange information to facilitate the transaction. For example, the mobile device, which may be the car itself, may transmit how much fuel is needed to fill the car up and credit card information to the pump. The pump may then authorize the dispensing of a preset amount of gas needed to fill the car up. Once the preset amount of gas is dispensed or the automatic shutoff of the pump activated, the POS may send a notification to the mobile device indicating that the transaction is complete with details of the transaction (i.e., amount of gas dispensed and total cost).

(11) Also, based on the flow rate of the pump and the desired amount of fuel needed, the pump may transmit an estimated time needed to fill up the car. As a result, the user may be able to better judge how long he or she may shop in the store without needlessly tying up a pump when the car is full. As disclosed herein, other POS systems besides fuel pumps may utilize beacons and cloud technology to improve payment efficiency.

(12) Turning now to the figures, FIG. 1 illustrates an example schematic of an operating environment **100** consistent with embodiments disclosed herein. As shown in FIG. 1, the operating environment **100** may include a POS system **102** and a mobile device **104** in communication with one another via a network **106**. Also as shown in FIG. 1, the POS system **102** may include a beacon **108**. While FIG. 1 shows a single POS system, multiple POS systems may be included in the operating environment **100**. For example, when the operating environment **100** is a gas station, POS system **102** may be a pump and there may be multiple pumps located at the station. If the operating environment is a traditional brick and mortar retailer, the POS system **102** may be a cash register or other terminal capable of processing payments.

(13) During operation, as the customer approaches the POS system **102**, the mobile device **104** may receive a beacon identifier transmitted by the POS system **102**. The beacon identifier may include data that distinctly identifies the POS system **102** from a plurality of POS systems. Upon receiving the beacon identifier, the mobile device **104** may transmit, via the network **106**, payment information to the POS system **102**. The payment information may include credit or debit card information, a desired quantity of a product, a maximum amount of money that may be spent for a product, etc. The desired quantity of the product and the maximum amount of money that may be spent for the product may be based on a price of the product.

(14) In addition, the POS system **102** may transmit, via the network **106**, product data or other information to the mobile device **104**. For example, the POS system **102** may transmit a price for a product, an estimate of a total purchase price, a time to complete a transaction, etc. Once a transaction has been completed, the POS system **102** may transmit a notification to the mobile device that the transaction is complete. For example, the POS system **102** may transmit an electronic receipt to the mobile device.

(15) As disclosed herein, the POS system **102** may be any type of system capable of processing a purchase transaction. Non-limiting examples of the POS system **102** include a cash register, a self-service terminal, a self-service kiosk, a fuel pump, etc. In addition to traditional POS systems, the POS system **102** may include a computing device coupled to a transmitter, receiver, or transceiver capable of exchanging information with the mobile device **104**, and not readily apparent or visible to the customer. For example, as disclosed herein, the POS system **102** may be a computing device located in a back area of a store (or even offsite) coupled to a transceiver located proximate an entrance to the store. As the mobile device **104** enters the store, a beacon or beacons located

proximate the entrance or throughout the store may send beacon identifiers to the mobile device **104**. Upon receiving the beacon identifiers, the mobile device may transmit a signal, via the network **106**, to the POS system **102** to begin a transaction or other exchange of product or payment data.

(16) The mobile device **104** may be any portable electronic device associated with a customer. Non-limiting examples of the mobile device **104** include a cellphone, smart phone, table computer, phablet, smart watch, smart car, etc. For instance, a car may have the capability to communicate with other devices via the Internet or other networks disclosed herein and have an onboard computing device capable of transmitting, receiving, and otherwise processing data. As such, the car may be considered a smart car. As the car approaches the POS system **102** (for instance, a fuel pump) the car and the POS system **102** may exchange information as disclosed herein.

(17) The network **106** may be any network capable of data transmission. Non-limiting examples of the network **106** include the Internet, cellular networks, peer-to-peer networks, wide area networks (WAN), local area networks (LAN), ad-hoc networks created by the mobile device **104** and the POS system **102**, or any combinations thereof. For example, the network may include a cellular network that a cellphone (i.e., the mobile device **104**) used to connect to the Internet, which connects to a LAN to which a cash register (i.e., the POS system **102**) is connected.

(18) The beacon **108** may be BLUETOOTH® beacons, WI-FI® hotspots, routers, etc. that allow the POS system **102** to broadcast the beacon identifier such that the mobile device may receive the beacon identifier. The power or range of the beacon **108** may depend on an application. For example, beacons associated with fuel pumps may have a short range (e.g., about 2 to 5 feet) and beacons for a brick and mortar retail store may have a longer range (e.g., about 10 to 20 feet or more). In addition, where the range of beacons may cause overlap of beacon identifiers, signal strength may be used to determine to which beacon the mobile device may be closer. For example, if the mobile device detects a first beacon identifier from a first beacon and a second beacon identifier from a second beacon, an app operating on the mobile device **104** may determine that the signal strength for the first beacon identifier is stronger than the signal strength for the second beacon identifier. Thus, the app may choose the first beacon as the closest beacon and initiate data exchange with the first beacon instead of the second beacon.

(19) Non-limiting examples of the operating environment **100** can include a gas station and a brick and mortar retail establishment. In a gas station context, the system may operate as follows. A car may pull up to a pump and the mobile device **104** (e.g., the car, the driver's cellphone or smart watch) may detect the beacon identifier emitted from the beacon **108**. Upon detecting the beacon identifier, the mobile device **104** may initiate communications with the POS system **102** via the network **106**. Once the connection is established, the POS system **102** may transmit price information about the various grades of gas offered. For example, the POS system **102** may transmit that regular unleaded is \$2.50/gallon and premium unleaded is \$2.80/gallon.

(20) Based on the price of the gas, the mobile device may transmit an amount of gas to be purchased. For example, if the price of regular unleaded is less than or equal to \$2.75/gallon the mobile device **104** may transmit that the pump is to dispense gas until the pump's automatic shutoff stops the flow of gas. In other words, the pump is to dispense gas until the car is full. If the price is above the preset value the mobile device **104** may transmit a request to purchase a preset amount of gas. For instance, if gas is above \$2.60/gallon the mobile device **104** may transmit that only \$20 worth of gas is to be dispensed. The automatic shutoff may override the preset amount for safety. For example, if the mobile device **104** authorizes \$20 worth of gas to be purchased, but the car will only hold an addition, \$10 worth of gas, once the automatic shutoff stops the flow of gas, the POS system **102** may complete the transaction as selling only \$10 worth of gas.

(21) Still consistent with embodiments disclosed herein, the mobile device **104** may authorize a purchase without specifying an amount. For example, the beacon **108** may periodically transmit the beacon identifier. In other words, the beacon **108** may periodically ping the mobile device **104**. The

mobile device **104** may receive the beacon identifier and transmit data to the POS system **102** indicating that the user is continuing a purchase. As such, the pump may continue to dispense fuel as long as the mobile device **104** continues to receive the beacon identifier and transmit data to the POS system **102**. Once the user is finished pumping gas and drives away, the mobile device **104** will no longer receive the period broadcast of the beacon identifier. After the loss of the beacon identifier reception, the mobile device **104** may transmit a data to the POS system **102** indicating that the transaction is complete. Once the transaction is complete, the POS system **102** may bill the user credit or debit card appropriately and transmit a push notification to the mobile device **104**. The push notification may be an electronic receipt for the fuel purchase and may contain data indicating how much gas was purchased, the total cost for the purchase, any loyalty rewards (include a rewards balance) the user may have earned, etc.

(22) As indicate herein, the car may be a smart car or otherwise coupled to a mobile device carried by the user. For example, the car may transmit data to the user's cellphone (e.g., via a Bluetooth® connection), which in turn may transmit data to the POS system **102**. The data transmitted by the car or cellphone may include how many gallons of gas are needed to fill the car up. For instance, the onboard computer of the car may determine that 7.5 gallons of gas are needed to fill the fuel tank. As such, the car may transmit to the user's cellphone (or directly to the POS system **102**) an authorization to purchase 7.5 gallons of gas.

(23) With a known quantity of gas to be dispensed and a known flowrate of the fuel pump, the POS system **102** may transmit to the mobile device **104** an estimated time to complete the transaction. For example, if the POS system **102** knows the pump dispenses gas at 2 gallons per minute and 7.5 gallons are needed, then the POS system **102** may transmit data to the mobile device **104** indicating that the transaction may take approximately 3.75 minutes to complete. In addition, calculating the approximate time to complete the transaction, the mobile device **104** may display a countdown timer that indicates the estimated time to complete the transaction.

(24) As indicated herein, the completion of the transaction may be indicated with the mobile device **104** leave a proximity to the POS system **102**. Once the transaction is complete in addition to transmitting a notification to the mobile device **104**, the POS system **102** may transmit a notification to a clerk or other attendant supervising the fuel pumps.

(25) Still consistent with embodiments disclosed herein, the operating environment **100** may be a traditional brick and mortar retail establishment. In this context, the mobile device **104** may be a user's cellphone or smartwatch. The beacon **108** may be located proximate a cash register or other check station. In addition, the beacon **108** may connect to shopping cart. As such, when the user selects a cart, the mobile device **104** may transmit data to the POS system **102** coupling the cart to the user. The cart may also include a radio frequency identification (RFID) scanner. As the user places items in the cart, the RFID scanner may detect a RFID tag associated with the product. The RFID scanner may be in electrical communication with the beacon **108** or other transmitter associate with the POS system **102**. As such, the RFID scanner or beacon **108** may transmit product data directly to the POS system **102** and the mobile device **104**.

(26) The product data may include a price for the product, expiration dates, budget categories, etc. An app running on the mobile device **104** may keep a running total of the user's purchases. The app may also include or interface with a budget program. As such, the app may alert the user the user if the user is spending more than the user has budgeted.

(27) The cart may also include a scale system that monitors the weight of items placed in the cart. For example, the legs or basket of the cart may include strain gauges that can measure determine a weight of products added to the cart. As such, the cart may determine a cost for products sold be weight. For instance, if a user place two pounds of bananas or grapes in the cart, the strain gauges may detect a strain, which is converted to a weight by the POS system **102**, the mobile device **104**, or the cart itself, to determine a total price for the bananas or grapes by multiplying the weight by the price/pound.

(28) The app running on the mobile device may include or interface with a shopping list app. As such, the user can create a shopping list while away from the store. While in the store, the mobile device **104** can use beacons located throughout the store to direct the user to the items on the shopping list. In addition, the app may suggest alternative items. For example, the app may suggest a store or generic product as opposed to a brand name to increase sales of the store brand and save the user money. The app may also alert the user to sales that may be of interest to the user based on the items in the shopping cart.

(29) The transaction can be completed when the mobile device **104** exits the store. POS system **102** or the mobile device **104** may determine that the mobile device **104** has exited the store in a variety of ways. For example, when the mobile device first initiates contact with the POS system **102**, the POS system **102** may transmit coordinate data for a geofence that surrounds the store. Once the mobile device **104** detects that it is outside the geofence, the mobile device **104** may transmit a message to the POS system **102** indicated that the transaction is to be completed. The POS system **102** may then transmit a notification or other electronic receipt to the mobile device **104** to indicate the transaction has been completed. In addition, just as with the gas station example, the completion of the transaction can be indicated when the mobile device **104** no longer receives a beacon identifier associated with the cart or the store.

(30) Regardless of the operating environment, retail store, gas station, etc. the systems and methods disclosed herein allow for transactions to occur automatically in the background without direct input from the user. As discussed herein, the systems and methods disclosed herein facilitate transactions without a consumer having to interact with a self-service terminal, cashier, or otherwise produce a tangible form of currency to complete a transaction. This represents a drastic improvement of current systems and methods for completing a transaction. The increase in efficiency by allowing consumers to autonomously complete transactions represents a huge advancement in the technology. This improves the operation of the hardware as well as uses the hardware in unconventional ways than previously imagined.

(31) For example, in the brick and mortar store example above, the systems and methods disclosed herein may eliminate checkout lines, and more importantly, the time wasted standing in them. In the gas station example, the systems and methods disclosed herein may shave 2 or more minutes off of each consumer's time at the pump. The 2 or more minutes may result in more consumers being able to purchase fuel, thereby increasing revenue for the station. For instance, if the average consumer spends 6 minutes at the pump, shortening that time by 2 minutes may allow an extra 5 customers to purchase fuel. For a station with 10 to 20 pumps that could mean an extra 50 to 100 customers per hour.

(32) FIG. 2 shows an example schematic of a computing device **200**. Computing device **200** may represent the POS system **102** or the mobile device **104**. As shown in FIG. 2, the computing device **200** may include a processor **202** and a memory unit **204**. The memory unit **204** may include a software module **206**, user data **208**, and product data **210**. While executing on the processor **202**, the software module **206** may perform processes for autonomously completing a transaction, including, for example, one or more stages included in method **300** described below with respect to FIG. 3.

(33) The software module **206** may include data or applications used to operate the computing device **200**. For example, if the computing device **200** is the POS system **102**, the software module **206** may include applications that operate components of the computing device **200** as disclosed herein. In addition, the software module **206** may include applications for transmitting the notifications, completing the transactions, updating an inventory management system etc. For example, when a transaction is completed, the computing device may transmit the sales data to an inventory management system to decrement the inventory of the items purchased.

(34) If the computing device **200** is the mobile device **104**, the software module **206** may include applications that operate components of the computing device **200** as disclosed herein. For

instance, the computing device **200** may execute application stored in the software module **206** to monitor for beacon signals, transmit user or payment information to the POS system **102**, display the notification upon completion of the transaction, etc.

(35) The user data **208** may include a user profile. The user profile may be created by a user of the mobile device **104**. For example, the user may create a profile that includes clothing sizes, billing information (e.g., credit card numbers and billing addresses), gas preferences (e.g., regular unleaded v. premium unleaded), maximum prices to pay for items (e.g., prices for gas that warrant filling the tank v. buying a preset amount in the hopes that prices drop), budgets, shopping lists, etc. The user profile may also include loyalty rewards information such as usernames and passwords, account balances, etc.

(36) The user profile may also be create by the POS system **102**. For example, the POS system may supplement the user profile created by the user to include additional information. For instance, the user profile may include spending trends, favorite brands, most commonly purchased items, etc. The user profile may be analyzed for marketing purposes or to better understand customer behaviors to improve customer service. The user profile may also include loyalty rewards information such as usernames and passwords, account balances, etc.

(37) The product data **210** may include prices for goods and services, sales data e sales trends), discounts, rewards credits for items, etc. The product data may also include information about the dispenser of the product. For example, the product data may include flowrate data for a gas pump, tare weights for produce containers, etc.

(38) The computing device **200** may also include a user interface **212**. The user interface **212** can include any number of devices that allow a user to interface with the computing device **200**. Non-limiting examples of the user interface **212** include a keypad, a microphone, a display (touchscreen or otherwise), etc.

(39) The computing device **200** may also include a location sensor **214**. The location sensor **214** can include any number of devices that provides information to allow the computing device **200** to determine its location. A non-limiting example of the location sensor **214** includes a GPS receiver, beacon receivers, etc.

(40) The computing device **200** may also include a communication port **216**. The communication port **216** may allow the computing device **200** to communicate with the beacon **108** and the network **106** as described herein. Non-limiting examples of the communication port **216** include, Ethernet cards (wireless or wired), BLUETOOTH® transmitters and receivers, near-field communications modules, cellular transmitters and receivers, etc.

(41) The computing device **200** may also include an input/output (UO) device **218**. The I/O device **218** may allow the computing device **200** to receive and output information. Non-limiting examples of the I/O device **218** include, a camera (still or video), a printer, a scanner, scales, RFID readers, etc.

(42) As disclosed herein, the computing device **200** may be implemented using a personal computer, a network computer, a mainframe, a handheld device, a personal digital assistant, a smartphone, smartwatch, tablet, phablet, or any other similar microcomputer-based workstation.

(43) FIG. 3 shows an example flowchart for a method **300** consistent with example embodiments disclosed herein. As shown in FIG. 3, the beacon **108** may broadcast a beacon identifier that may be received by the mobile device **104** (line **302**). Once the mobile device **104** detects the beacon **108**, the mobile device **104** may transmit user data and payment information authorizing payment for a product via the network **106** to the POS system **102** (lines **304** and **306**). As disclosed herein, the user data and payment information may be how much as is to be purchased, a request for product information, a shopping lists, etc.

(44) Prior to or after, depending on the configuration of the system and the operating environment, the POS system **102** may transmit product data, via the network **106**, to the mobile device (lined **308** and **310**). For example, in a gas station setting, the POS system **102** may push price data to the

mobile device **104** prior to the mobile device authorizing a payment. Still consistent with a gas station setting, upon detecting the beacon **108**, the mobile device **104** may authorize payment for any amount of gas without regard to the price. In a retail store setting, as a user places a product in the cart, the mobile device may request product information (e.g., price). Only after the user has exited the store, or reached a predetermined spot within the store that indicates the user is done shopping, may the mobile device **104** transmit the payment authorization.

(45) Once payment has been authorized, the POS system **102** may transmit data, via the network **106**, to the mobile device **104** indicating that dispensing can occur and how much is being dispensed (lines **312** and **314**). For example, in the gas station setting, the pump may transmit an indication that the pump is dispensing gas and the rate it is being dispensed to the user's cellphone or directly to the car itself (if the car is the mobile device **104**). If the user's cellphone is the mobile device **104**, the cellphone may further transmit data to the car (via Bluetooth® or other near-field-communication protocol) indicating how much gas is being deposited in the gas tank.

(46) The mobile device **104** may lose the ability to detect the beacon **108**. For example, the stages represented by lines **302**, **304**, and **306** may periodically repeat. When one of those stages is interrupted (i.e., the mobile device **104** is no longer proximate the beacon **108**), the POS system **102** may complete the transaction by charging the user's payment method and send a notification, via the network **106**, to the mobile device **104** as disclosed herein indicating the transaction is complete (lines **314** and **318**).

(47) In the alternative, the mobile device **104** may transmit, via the network **106**, a signal to the POS system **102** indicating the transaction is to be completed (lines **320** and **322**). Once the POS system **102** completes the transaction, the POS system **102** may transmit, via the network **106**, a notification or electronic receipt to the mobile device **104** indicating that the transaction is complete (lines **324** and **326**).

(48) Additional examples and embodiments:

(49) Example 1 is a method comprising: transmitting, by a beacon associated with a point-of-sale system, a beacon identifier receivable by a mobile device; receiving, by the point-of-sale system and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for a product.

(50) In Example 2, the subject matter of Example 1 optionally includes wherein the point-of-sale system is a fuel pump.

(51) In Example 3, the subject matter of any one or more of Examples 1-2 optionally include wherein the product is gasoline or diesel.

(52) In Example 4, the subject matter of any one or more of Examples 1-3 optionally include wherein the mobile device is a smart phone, a smart watch, or a smart automobile.

(53) In Example 5, the subject matter of any one or more of Examples 1-4 optionally include transmitting, by the point-of-sale system, a price of the product to the mobile device.

(54) In Example 6, the subject matter of Example 5 optionally includes wherein receiving the payment information includes receiving a desired quantity of the product to be purchased based on the price of the product.

(55) In Example 7, the subject matter of any one or more of Examples 1-6 optionally include transmitting, by the point-of-sale system, a push notification to the mobile device, the push notification indicating an amount of the product purchased and that the purchase is complete.

(56) In Example 8, the subject matter of Example 7 optionally includes periodically receiving a ping acknowledgement from the mobile device once the mobile device receives the beacon identifier, wherein the push notification is transmitted to the mobile device when the point-of-sale system does not receive the ping acknowledgement from the mobile device within a preset time interval.

(57) Example 9 is a method comprising: receiving, at a mobile device from a beacon, a beacon identifier from a beacon associated with a point-of-sale system; and transmitting, by the mobile

device and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for a product.

(58) In Example 10, the subject matter of Example 9 optionally includes wherein the point-of-sale system is a fuel pump.

(59) In Example 11, the subject matter of any one or more of Examples 9-10 optionally include wherein the product is gasoline or diesel.

(60) In Example 12, the subject matter of any one or more of Examples 9-11 optionally include wherein the mobile device is a smart phone, a smart watch, or a smart automobile.

(61) In Example 13, the subject matter of any one or more of Examples 9-12 optionally include receiving, at the mobile device, a price of the product.

(62) In Example 14, the subject matter of Example 13 optionally includes wherein transmitting the payment information includes authorizing a fixed amount of the product to be purchased based on the price of the product.

(63) In Example 15, the subject matter of Example 14 optionally includes wherein the fixed amount of the product to be purchased is transmitted to the point-of-sale system.

(64) In Example 16, the subject matter of any one or more of Examples 9-15 optionally include receiving, at the mobile device, a push notification indicating an amount of the product purchased and that the purchase is complete.

(65) Example 17 is a system comprising: a beacon; a receiver; a processor in electrical communication with the transmitter and the receiver; and a memory that stores instructions that, when executed by the processor, cause the processor to: transmit, via the beacon, a beacon identifier receivable by a mobile device, and receiving, via the receiver and without interaction by a user of the mobile device, payment information associated with the user, the payment information authorizing payment for a product.

(66) In Example 18, the subject matter of Example 17 optionally includes wherein the system is a component of a fuel pump and the product is gasoline or diesel.

(67) In Example 19, the subject matter of any one or more of Examples 17-18 optionally include wherein the operations further cause the processor to: transmit, via a transmitter, a price of the product to the mobile device; and receive a desired quantity of the product based on the price of the product.

(68) In Example 20, the subject matter of any one or more of Examples 17-19 optionally include wherein the operations further cause the processor to: periodically receiving a ping acknowledgment from the mobile device once the mobile device receives the beacon identifier; and transmit a push notification to the mobile device, the push notification indicating an amount of the product purchased and that the purchase is complete, wherein the push notification is transmitted to the mobile device when the point-of-sale system does not receive the ping acknowledgement from the mobile device within a preset time interval.

(69) It will be readily understood to those skilled in the art that various other changes in the details, material, and arrangements of the parts and method stages which have been described and illustrated in order to explain the nature of the inventive subject matter may be made without departing from the principles and scope of the inventive subject matter as expressed in the subjoined claims.

Claims

1. A method comprising: transmitting, by a beacon associated with a point-of-sale system, a beacon identifier to a mobile device in proximity to the point-of-sale system, wherein the point-of-sale system is associated with a fuel pump; receiving, by the point-of-sale system and without interaction by a user of the mobile device, payment information associated with the user, wherein the payment information authorizes a payment for a product wherein the payment information is

received by the mobile device from an onboard computer of an automobile, via a first wireless connection, and wherein the mobile device sends the payment information to the point-of-sale system, via a second wireless connection, without the user interacting with the onboard computer or without interacting with the mobile device; transmitting, by the point-of-sale system, a push notification to the mobile device, the push notification indicating an amount of the product purchased and indicating that a purchase is complete; transmitting, by the point-of-sale system to the mobile device, an estimated time to complete a transaction associated with the payment for the product, the estimated time being a function of a flow rate of the fuel pump and the amount of fuel needed; and transmitting, by the point-of-sale system, a notification to a clerk supervising the point-of-sale system and the mobile device, the notification indicating the transaction is complete, wherein the notification is presented on a display visible to the clerk.

2. The method of claim 1, wherein the product is gasoline or diesel.

3. The method of claim 1, wherein the mobile device is a smart phone, a smart watch, or a smart automobile.

4. The method of claim 1, further comprising transmitting, by the point-of-sale system, a price of the product to the mobile device.

5. A system comprising: a beacon; a receiver; a transmitter; a processor in electrical communication with the beacon, the transmitter, and the receiver; and a memory that stores instructions that, when executed by the processor, cause the processor to perform operations comprising: transmitting, via the beacon, a beacon identifier receivable to a mobile device in proximity to a point-of-sale system associated with the system, wherein the point-of-sale system is associated with a fuel pump, receiving, via the receiver and without interaction by a user of the mobile device, payment information associated with the user, wherein the payment information authorizes a payment for a product, wherein the product being an amount of fuel needed determined by an onboard computer of an automobile and communicated by the onboard computer, via a first wireless connection, to the mobile device and from the mobile device communicated, via a second wireless connection, to the point-of-sale system without the user interacting with the onboard computer or without the user interacting with the mobile device; transmitting a push notification to the mobile device, the push notification indicating an amount of the product purchased and indicating that a purchase is complete; and transmitting, by the transmitter, a notification to a clerk supervising the point-of-sale system and the mobile device, the notification indicating a transaction is complete, wherein the notification is presented on a display visible to the clerk.
